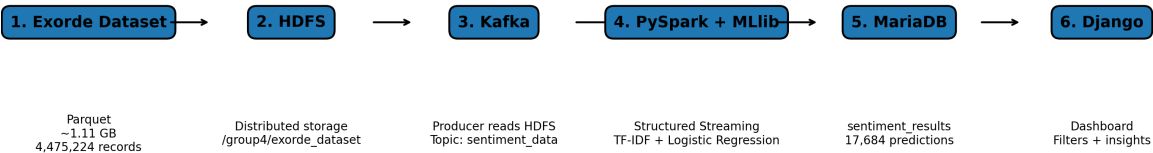


SOCIAL MEDIA SENTIMENT ANALYSIS

FINAL PROJECT DOCUMENTATION

Exorde Dataset → HDFS → Kafka → PySpark Structured Streaming → MLlib → MariaDB → Django

Group 4 - End-to-End Big Data Sentiment Analysis Architecture



HDFS → Kafka → PySpark Structured Streaming → MLlib → MariaDB → Django Dashboard

Figure 1. Implemented end-to-end project architecture.

GROUP 4 • October 2026

Group Members

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Submitted as the final course project for Big Data Essentials.

Executive Summary

This project presents an end-to-end Big Data pipeline for Social Media Sentiment Analysis. The implementation integrates Hadoop HDFS for large-scale storage, Apache Kafka for streaming, PySpark Structured Streaming for processing, Spark MLlib for machine-learning prediction, MariaDB for storing prediction results, and Django for web-based visualization.

The Exorde dataset contained 4,475,224 social-media records and approximately 1.11 GB of Parquet data across nine files. Historical data was stored in HDFS. The Kafka Producer read data from HDFS and published records to the sentiment_data topic. PySpark Structured Streaming consumed the Kafka stream and successfully detected newly arriving records.

A 2% sample was used for the machine-learning experiment because of local computing constraints. This produced 89,590 records, with 71,906 training records and 17,684 test records. The Spark MLlib Logistic Regression model achieved 69.45% accuracy, 69.20% precision, 69.45% recall, and 69.19% F1-score.

The 17,684 test prediction results were stored in MariaDB. The Django dashboard successfully displayed totals, sentiment percentages, model accuracy, recent predictions, and an interactive Positive/Negative/All filter.

Contents

1. Project Overview
 2. Project Objectives
 3. System Architecture
 4. Technologies Used
 5. Version Information
 6. Dependencies and Configuration
 7. Implementation Workflow
 8. Machine Learning Methodology
 9. Results and Visualizations
 10. Django Web Dashboard
 11. Testing and Validation
 12. Findings and Discussion
 13. Limitations and Future Improvements
 14. Conclusion
 15. References and Project Evidence
- Appendix A. Key Project Evidence

1. Project Overview

The project addresses the problem of extracting sentiment information from large-scale social media data using Big Data technologies. The objective is to demonstrate how historical data storage, real-time streaming, distributed processing, machine learning, database storage, and web visualization can be integrated into a single practical system.

1.1 Case Study

Social Media Sentiment Analysis.

1.2 Primary Data Source

Exorde social-media dataset in Parquet format.

1.3 Main Data Flow

Exorde Dataset → HDFS → Kafka → PySpark Structured Streaming → MLlib → MariaDB → Django Dashboard.

1.4 Problem Statement

Large volumes of social media posts contain useful opinions and reactions, but manually analyzing these posts is slow and difficult at scale. This project applies Big Data technologies to automate sentiment processing and make the resulting insights easier to inspect.

2. Project Objectives

- Store a large social-media dataset in HDFS.
- Use Kafka to simulate real-time streaming from HDFS.
- Consume and process streaming records using PySpark Structured Streaming.
- Apply data cleaning and text feature engineering for sentiment analysis.
- Train and evaluate a sentiment prediction model using Spark MLlib.
- Store prediction results in a MySQL-compatible database.
- Develop a Django dashboard for sentiment statistics and prediction results.
- Test the complete integrated pipeline.

3. System Architecture

The architecture follows the lecturer's required workflow. HDFS provides storage; the Kafka Producer reads records from HDFS and publishes them to a Kafka topic; PySpark Structured Streaming consumes and processes the stream; MLlib provides sentiment prediction; MariaDB stores the results; and Django provides the presentation layer.

Group 4 - End-to-End Big Data Sentiment Analysis Architecture

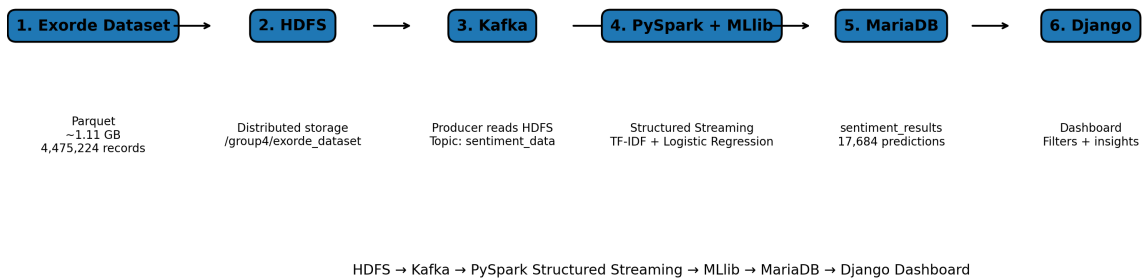


Figure 2. Group 4 end-to-end architecture.

3.1 Component Roles

Component	Role
Exorde Dataset	Source of social media posts and metadata.
HDFS	Stores the large Parquet dataset for Big Data processing.
Apache Kafka	Provides the message-streaming layer through sentiment_data.
PySpark Structured Streaming	Consumes Kafka messages, parses records, cleans data, and prepares features.
Spark MLlib	Trains and evaluates the Logistic Regression sentiment model.
MariaDB/MySQL	Stores final prediction results in sentiment_results.
Django	Provides the web dashboard for statistics, predictions, and filtering.

4. Technologies Used

Technology	Purpose
Hadoop HDFS	Large-scale distributed storage of the Exorde Parquet dataset.
Apache Kafka	Message broker and real-time streaming layer.
Apache Spark / PySpark	Distributed processing and Structured Streaming.
Spark MLlib	Machine-learning model training and evaluation.
MariaDB / MySQL	Storage of processed sentiment predictions.
Django	Web framework used for the project dashboard.
Python / Jupyter	Project scripting, experimentation, orchestration, and analysis.
PyArrow / Parquet	Reading and handling the columnar dataset format.
Matplotlib	Analytical and model-result visualizations.

5. Version Information

Tool / Library	Version
Hadoop	3.4.2
Apache Kafka	3.8.0
PySpark / Apache Spark	4.2.0
Python	3.12.14
Java / JDK	17.0.5
Django	4.2.27
MariaDB	10.4.32
mysqlclient	2.3.0
kafka-python-ng	2.2.3
PyArrow	25.0.1

The Spark/Kafka connector artifacts used in the project were built for Scala binary version 2.13. No separate Scala runtime installation was required for the notebook workflow.

6. Dependencies and Configuration

6.1 Python Dependencies

- pyspark 4.2.0
- kafka-python-ng 2.2.3
- PyArrow 25.0.1
- mysqlclient 2.3.0

- mysql-connector-python (database verification)
- Django 4.2.27
- Matplotlib

6.2 Spark/Kafka JAR Dependencies

- spark-sql-kafka-0-10_2.13-4.2.0.jar
- kafka-clients-3.9.2.jar
- spark-token-provider-kafka-0-10_2.13-4.2.0.jar
- scala-parallel-collections_2.13-1.2.0.jar
- jsr305-3.0.0.jar
- commons-pool2-2.13.1.jar

6.3 Project Configuration

- HDFS dataset path: /group4/exorde_dataset
- Kafka broker: localhost:9092
- Kafka topic: sentiment_data
- Database: sentiment_analysis
- Database table: sentiment_results
- Django application: localhost:8000

7. Implementation Workflow

Stage	Implementation
Step 1 – Dataset and HDFS	Exorde dataset verified locally, split into nine Parquet files, and uploaded to HDFS.
Step 2 – Kafka	Kafka Producer read a Parquet file from HDFS and published records to sentiment_data.
Step 3 – PySpark Streaming	Structured Streaming consumed Kafka records, parsed the schema, cleaned records, and prepared features.
Step 4 – MLlib	Historical data was used to train and evaluate Logistic Regression.
Step 5 – MariaDB	17,684 test prediction records were stored in sentiment_results.
Step 6 – Django	Django connected to the database and displayed statistics, predictions, accuracy, and filtering.
Step 7 – Testing	Django checks, database checks, dashboard tests, and deployment checks were performed.

8. Machine Learning Methodology

Sentiment values below zero were labeled Negative, while values equal to or above zero were labeled Positive. Text was tokenized and transformed into numerical features using Spark ML feature processing, including TF-IDF-related feature engineering. Logistic Regression from Spark MLlib was then trained and evaluated.

Item	Value
Full dataset	4,475,224
ML sample	89,590
Training records	71,906
Testing records	17,684
Model	Logistic Regression

9. Results and Visualizations

The complete dataset contained 4,475,224 records. Positive sentiment accounted for 2,564,821 records (57.31%), while Negative sentiment accounted for 1,910,403 records (42.69%).

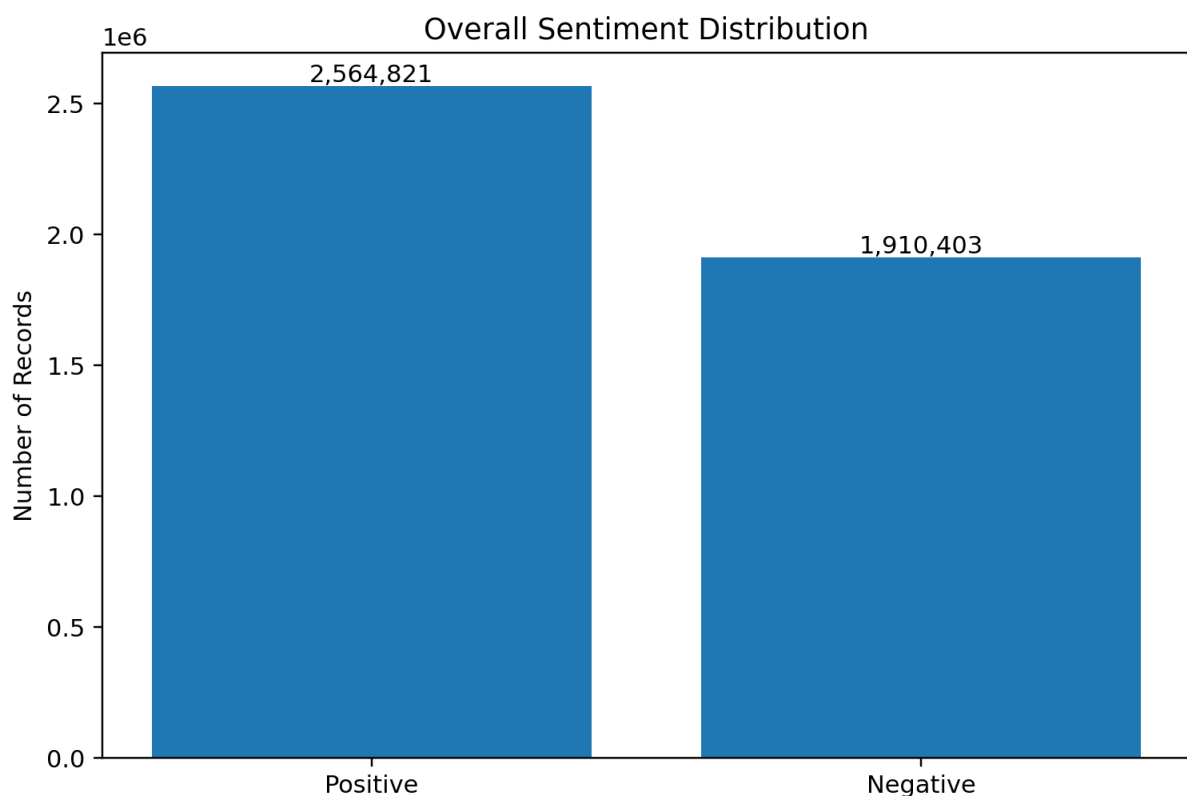


Figure 3. Overall sentiment distribution generated during Python analysis.

The machine-learning evaluation produced 69.45% accuracy, 69.20% precision, 69.45% recall, and 69.19% F1-score.

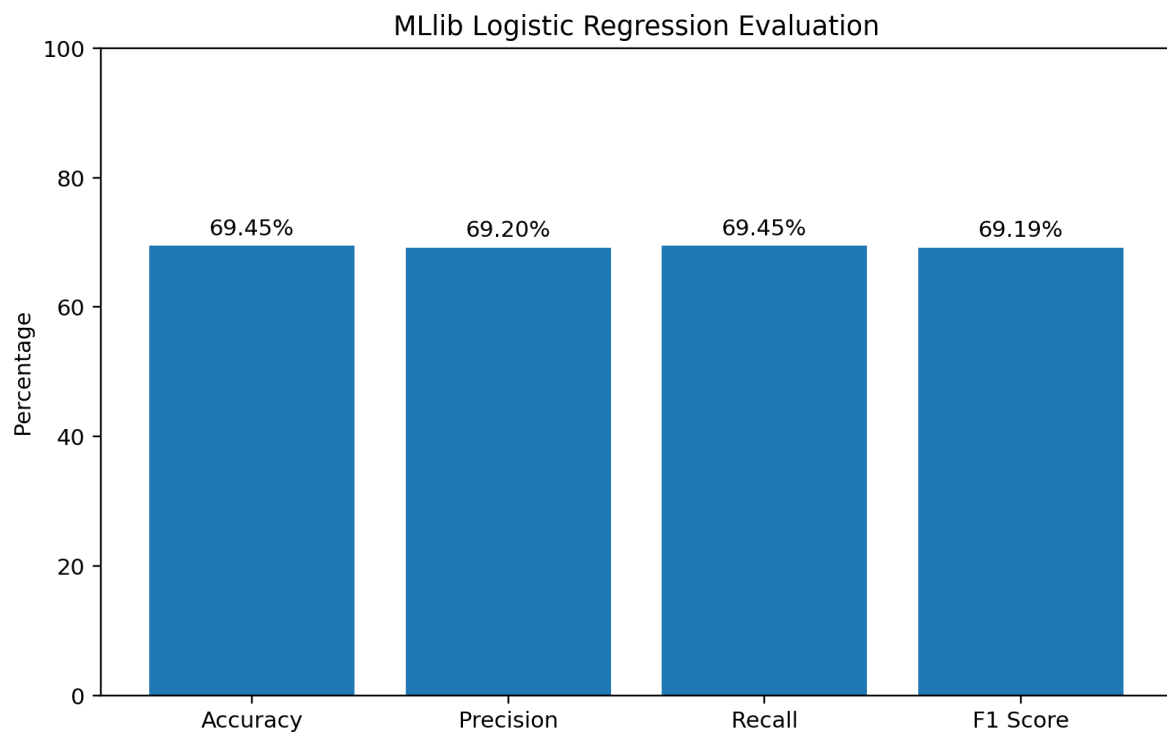


Figure 4. MLlib Logistic Regression evaluation metrics.

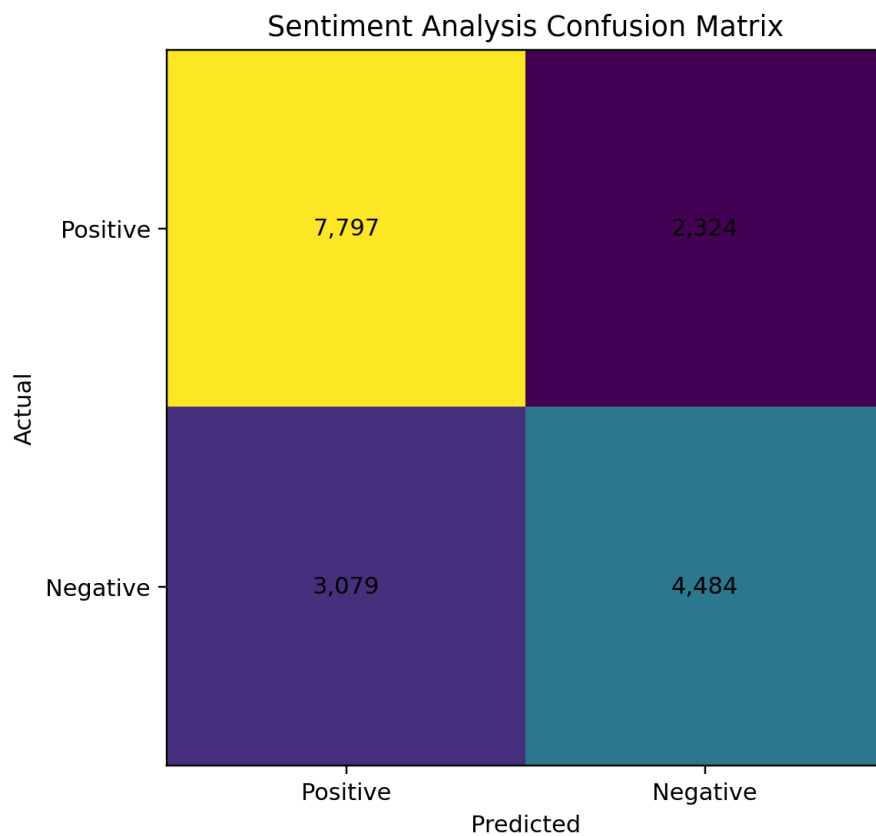


Figure 5. Confusion matrix for the held-out test set.

The confusion matrix contains 7,797 correctly predicted Positive records and 4,484 correctly predicted Negative records. There were 2,324 Positive records predicted as Negative and 3,079 Negative records predicted as Positive.

10. Django Web Dashboard

The Django dashboard provides the presentation layer of the system. It connects to the existing MariaDB sentiment_results table and displays aggregate results and individual predictions.

- 17,684 total stored prediction records.
- 10,876 Positive predictions and 6,808 Negative predictions.
- 61.5% Positive and 38.5% Negative prediction distribution.
- 69.45% model accuracy.
- Recent social-media prediction records.
- Interactive Positive / Negative / All filtering.

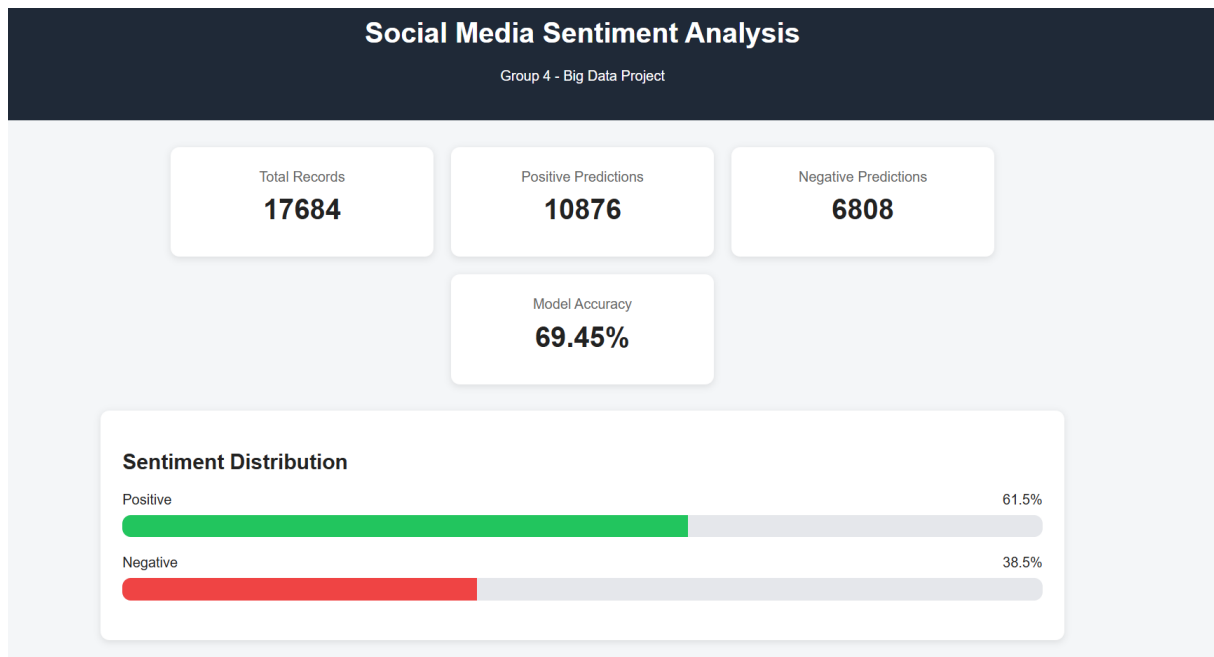


Figure 6. Django dashboard showing project results and model accuracy.

11. Testing and Validation

Test	Evidence	Status
HDFS dataset	Nine Parquet files uploaded and verified.	Passed
Kafka Producer	Read dataset data from HDFS and sent records.	Passed
Kafka Consumer	Kafka messages successfully read and verified.	Passed
Structured Streaming	Initial batch and newly arriving message processed.	Passed
MLlib model	Predictions generated and evaluated on 17,684 test records.	Passed
MariaDB	Connection succeeded and 17,684 records were stored.	Passed
Django model	Django queried 17,684 records from the existing table.	Passed
Dashboard	Dashboard loaded and displayed statistics and predictions.	Passed
Filter	Positive, Negative, and All filters tested successfully.	Passed
Django system check	python manage.py check returned no issues.	Passed

Test	Evidence	Status
Deployment check	Security warnings appeared because the project is local HTTP.	Passed with warnings

12. Findings and Discussion

- The complete dataset contained 4,475,224 records, demonstrating a sufficiently large dataset for the Big Data workflow.
- Positive sentiment was more common than Negative sentiment in the complete dataset, at 57.31% versus 42.69%.
- The Kafka pipeline successfully demonstrated movement of records from HDFS into a streaming topic.
- PySpark Structured Streaming successfully detected and processed newly arriving Kafka data.
- The Logistic Regression model achieved 69.45% accuracy on the held-out test set.
- Precision, recall, and F1-score were all close to 69%, indicating reasonably balanced overall predictive performance.
- The confusion matrix shows that the model identified both classes but still produced false-positive and false-negative errors.
- MariaDB and Django transformed analytical outputs into a usable presentation layer.

12.1 Interpretation of Model Performance

An accuracy of 69.45% means that the model correctly classified approximately seven out of every ten records in the test set. The remaining errors show that social-media language can be difficult to classify because posts may contain sarcasm, mixed opinions, short context, slang, or ambiguous wording. The model is therefore a practical baseline rather than perfect sentiment understanding.

13. Limitations and Future Improvements

- The machine-learning experiment used a 2% sample of the full dataset because of local computing constraints.
- The project was deployed locally rather than on a public production server.
- MariaDB was used locally instead of AWS RDS.
- A production version could connect the streaming prediction stage directly to the database for continuous dashboard updates.
- Model performance could be improved using stronger text normalization, stop-word handling, n-grams, richer embeddings, or alternative Spark MLlib classifiers.
- The dashboard could be extended with theme-level charts, time-based sentiment trends, pagination, downloadable reports, and live refresh.

14. Conclusion

The Social Media Sentiment Analysis project successfully demonstrated the integration of the major technologies required for the Big Data Essentials course. The implementation moved from large-scale HDFS storage to Kafka streaming, PySpark Structured Streaming, Spark MLlib machine learning, MariaDB storage, and Django visualization.

The final system was tested component by component and as an integrated application. The Logistic Regression model achieved 69.45% accuracy, while the Django dashboard successfully displayed 17,684 stored prediction records, sentiment percentages, model accuracy, recent predictions, and interactive filtering.

Overall, the project demonstrates a complete practical Big Data workflow and provides a foundation that can be extended toward continuous production-scale sentiment monitoring.

15. References and Project Evidence

1. Big Data Essentials Course Project Guidelines for Students, 2026 — lecturer-provided project requirements.
2. Group 4 – Social Media Sentiment Analysis Final Notebook, Group4_Sentiment_Analysis_Final.ipynb.

3. Project implementation environment and locally configured Hadoop, Kafka, Spark, MariaDB, and Django services.
4. Exorde social-media dataset in Parquet format used as the primary project dataset.

Appendix A. Key Project Evidence

Evidence	Value
HDFS dataset path	/group4/exorde_dataset
Kafka topic	sentiment_data
Kafka broker	localhost:9092
Django application	localhost:8000
Database	sentiment_analysis
Database table	sentiment_results
Database records	17,684
Full dataset records	4,475,224
ML sample	89,590
ML training records	71,906
ML test records	17,684
Model accuracy	69.45%
Positive prediction records	10,876
Negative prediction records	6,808

END OF FINAL PROJECT DOCUMENTATION